

Project	In-car	Released version: 2015-07-15
Parts for family qualification	SLI97C(U)(N)(F)(X)xxxxPE	
Chip for family qualification	M9905	
Device family	SLI 97	

Part Operating Temperature Grade

2; -40°C to 105°C

Family qualification (generic data) with structural similar (representative) types

	Part	Chip	Chip size	Wafer diameter	Wafer fab	Wafer technology	Package	Assembly line
Part to be Qualified:	SLI 97CFxxxxPE	M9905	2.762 mm x 2.592 mm	200 mm	IFX DD	L90FL	PG-VQFN-8-4	Amkor Philippines
1. Mechanical representative (generic family part A):	SLI 76CF5120P	M7741	2.88 mm x 2.08 mm	200 mm	IFX DD	C11FL	PG-VQFN-8-1	Amkor Philippines
2. Mechanical representative (generic family part B):	SLM 76CF5120P	M7711	2.88 mm x 2.08 mm	200 mm	IFX DD	C11FL	PG-VQFN-8-1	Amkor Philippines
3. Mechanical representative (generic family part C):								

Explanation:

Mechanical representative type: Same or bigger chip size, Differences max. 1.5x and 1.5y
Same package technology
Same die bond method (glue or solder die bond)
Same wire bond method (Au nail head or Al wedge)

Electrical representative type: Same wafer technology / wafer process
Same wafer fab
Same wafer diameter

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TEST #	STRESS TEST according AEC Q100 Rev. G	Test conditions	# LOTS	Sample Size	Part to be Qualified	Generic Family part A	Generic Family part B	Generic Family part C	Remarks
TEST GROUP A – ACCELERATED ENVIRONMENT STRESS TESTS									
A1	Preconditioning	MSL: 1 peak temp.: 260°C J-STD-020, JESD22-A113	3	231	PASS				
A2	Temperature Humidity Bias	85°C/85%RH/1000h JESD22-A101	3	77	PASS				
A3	uHAST	JESD22-A118 130°C / 85% r.h. 230kPa 96h	3	77	PASS				
A4	Temperature Cycling	-50/150°C, 500x JESD22-A104	3	77	PASS				
A5	Power Temperature Cycling	n.a.			not applicable				SLI 97 is no power chip, rated < 1W
A6	High Temperature Storage Life	150°C, 500h JESD22-A103	1	45	PASS				
TEST GROUP B – ACCELERATED LIFETIME SIMULATION TESTS									
B1	High Temperature Operating Life	125°C, 1000h JESD22-A108	3	77	PASS				
B2	Early Life Failure Rate	125°C AEC Q100-008	3	800	PASS				
B3	NVM Endurance, Data Retention, & Operational Life	P/E cycles: 100k and 500k HTSL: 150°C/500h HTOL: 125°C/408h AEC Q100-005	3	77	PASS				
TEST GROUP C – PACKAGE ASSEMBLY INTEGRITY TESTS									
C1	Wire Bond Shear	AEC Q100-001	1	6	PASS				after TC
C2	Wire Bond Pull	MIL-STD-883 - 2011	1	6	PASS				after TC
C3	Solderability	JESD22-B102	1	15	PASS				
C4	Package Dimensions	JESD22-B100/108 acc. package specifications	3	10	PASS				for results see separate report
C5	Solder Ball Shear				not applicable				No solder balls on VQFN8
C6	Lead Integrity				not applicable				No leads on VQFN8

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TEST GROUP D – DIE FABRICATION RELIABILITY TESTS									
D1	Electromigration				PASS				PASS, part of technology qualification
D2	Time Dependent Dielectric Breakdown				PASS				PASS, part of technology qualification
D3	Hot Carrier Injection				PASS				PASS, part of technology qualification
D4	Negative Bias Temperature Instability				PASS				PASS, part of technology qualification
D5	Stress Migration				PASS				PASS, part of technology qualification
TEST GROUP E – ELECTRICAL VERIFICATION TESTS									
E1	Pre- and Post-Stress Function Parameter	Test @ 25°C, 105°C	all	all	PASS				
E2	Electrostatic Discharge Human Body Model	2kV (all interfaces) AECQ100-002	1	3/voltage	PASS				
E3	Electrostatic Discharge Charged Device Model	750V AECQ100-011	1	3/voltage	PASS				
E4	Latch-Up	±150mA, 105°C JESD78	1	6	PASS				
E5	Electrical Distributions	AEC Q100-009	3	30	PASS				
E6	Fault Grading	AEC-Q100-007							see separate document
E7	Characterization	AEC Q003	3	10	PASS				
E9	Electromagnetic Compatibility	based on SAE J1752/3	1	1	PASS				IFX internal procedure, aligned for GSM operation
E10	Short Circuit Characterization				not applicable				SLI 97 is no smart-power device
E11	Soft Error Rate				not applicable				SLI 97 contains SRAM < 1MByte
TEST GROUP F – DEFECT SCREENING TESTS									
F1	Process Average Testing	AEC Q001							Iddq not available by design implemented for current modes
F2	Statistical Bin/Yield Analysis	AEC Q002							implemented for all softbins



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TEST GROUP G – CAVITY PACKAGE INTEGRITY TESTS									
G1	Mechanical Shock	JESD22-B104 5 pulses, 0.5 ms, 1500g	3	26	not applicable	PASS			For cavity packages only As information for ETSI compliance
G2	Variable Frequency Vibration	JESD22-B103 20-2000Hz	1	39	not applicable		PASS		For cavity packages only As information for ETSI compliance
G3	Constant Acceleration				not applicable				For cavity packages only
G4	Gross/Fine Leak				not applicable				For cavity packages only
G5	Package Drop				not applicable				For cavity packages only
G6	Lid Torque				not applicable				For cavity packages only
G7	Die Shear Strength				not applicable				For cavity packages only
G8	Internal Water Vapor				not applicable				For cavity packages only